UNIVERSITY OF MICHIGAN · IOE 562 · WINTER 2026

Hourly Electricity Demand Forecasting for Toledo

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A three-approach comparison: OLS diagnostics / estimator benchmarking / deep learning

Why hourly load forecasting matters

Utilities forecast electricity load at every horizon, from 5 minutes to 20 years. We study the hourly slice. It is the same resolution that drives day-ahead dispatch, monthly bill estimates, renewable scheduling, and year-ahead capacity auctions.

Grid operators [1]

24 hourly models per PJM zone

PJM ATSI (which serves Toledo Edison) commits capacity worth $333.44/MW-day in the 2027/28 auction. Day-ahead forecasts refresh every 6 hours.

Consumers [2]

~300,000 customers served

Toledo Edison: 91.7% residential, 8.25% commercial, 0.08% industrial. A 1% monthly-bill error on the ~12¢/kWh residential rate is ~$9 per customer per year.

Renewables (Ohio 2024 mix) [4]

5.2% renewables · 59.8% gas · 21.1% coal

Solar 2.8% · Wind 2.0% · Hydro 0.4% · Biomass 0.2%. Nuclear 12.6% (one reactor near Toledo). Variable renewables need an accurate load curve to schedule storage and backup gas.

[1] PJM Manual 19: Load Forecasting (pjm.com/library/manuals) [2] FirstEnergy · firstenergycorp.com/toledo_edison.html [3] PJM 2025 Long-Term Load Forecast Report (pjm.com/library/reports-notices/load-forecast) [4] EIA Ohio State Energy Profile 2024 · eia.gov/state/?sid=OH

Data: sources, cleaning, COVID decision

LOAD

Toledo Edison (FirstEnergy)

Hourly load by customer class (Residential · Commercial · Industrial).
Non-Shopped + Shopped + PIPP. Source: Ohio PUCO filings.
Jan 2019 – Feb 2026. 62,784 raw rows.

WEATHER

Open-Meteo reanalysis

19 hourly variables: temperature, apparent temp, humidity, wind, gusts, precipitation, shortwave / direct / diffuse radiation, VPD, soil.
Toledo grid cell. Jan 2019 – Dec 2025. 61,368 rows. Zero missing values.

Cleaning (minimal — file was well-maintained)

1. Mixed datetime formats → POSIXct
2. Hour 24 → midnight of next day
3. Comma-formatted numbers (e.g. 1,234,567) → numeric
4. UTC weather → Eastern time; DST duplicates dropped

COVID window (Mar 2020 – Jun 2021): we tried both

A) Add a binary COVID flag alongside the other indicators. B) Drop the 10,992 lockdown rows entirely before fitting. Final models use (B). Lockdown demand is a policy regime, not a function of weather or calendar, and leaving it in forces every coefficient to compromise.

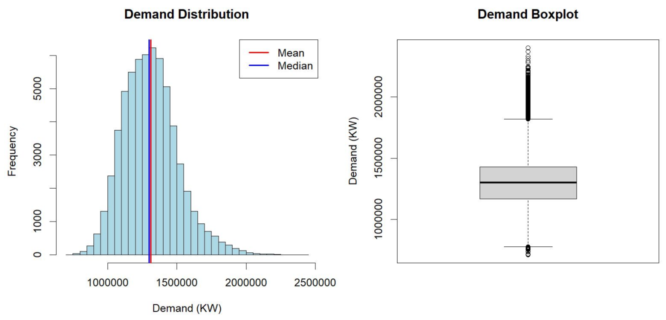
Final modeling dataset

50,235 hourly rows · 21 columns · no interpolation applied · range 710,935 – 2,400,995 KW · mean 1,310,413 KW

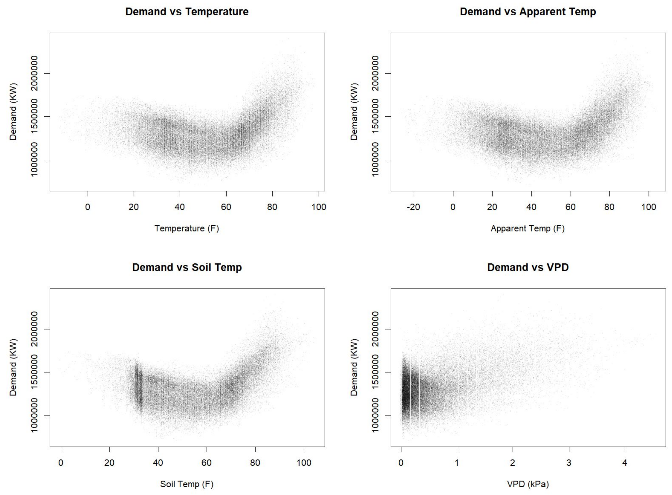
Scope: Toledo Edison only (one utility, one city, one weather cell). Geographic match to a single reanalysis point.

EDA: what the data told us to do

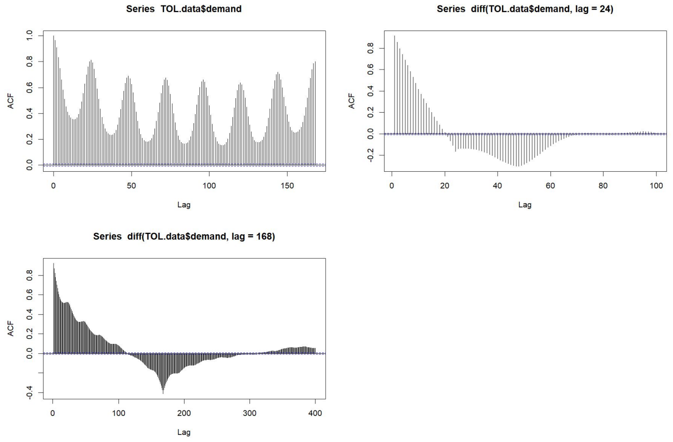
Distribution



Demand vs apparent temperature



Autocorrelation of demand



OBSERVED

Bimodal, right-skewed. Winter + summer peaks.

INTERPRETED

Two HVAC regimes (heating + cooling). A linear temp slope cannot capture both arms.

ACTION TAKEN

Box-Cox log response; residual variance no longer fans out.

OBSERVED

Asymmetric U-shape. Cooling arm steeper than heating arm.

INTERPRETED

Temperature effect is strongly nonlinear. Minimum near 55-60°F.

ACTION TAKEN

A1: B-spline knots at 30, 50, 70°F. A2: cubic polynomial on apparent_temp.

OBSERVED

Sharp ACF peaks at lag 24 and lag 168 (hours).

INTERPRETED

Daily and weekly cycles dominate. OLS residuals stay autocorrelated.

ACTION TAKEN

A1: +Lag-1 term. A2: +Lag-24, +Lag-168. A3 LSTM: sequence length 24.

Baseline: what we are trying to beat

RESPONSE & PREDICTORS

Response: log(demand), hourly KW

Predictors used across baselines: 19 weather variables (temperature, apparent temp, humidity, wind, gusts, precipitation, shortwave / direct / diffuse radiation, VPD, soil temp, soil moisture) + hour-of-day, day-of-week, month, season factors + holiday and heatwave flags.

OLS · weather + calendar (general regression baseline)

All 19 weather predictors + hour / dow / season factors. No transform, no lag.

σ ≈ 106,476 KW | adj R^2 ≈ 0.71

SARIMA(1,0,3)(2,1,0)[24] · pure time series, no weather

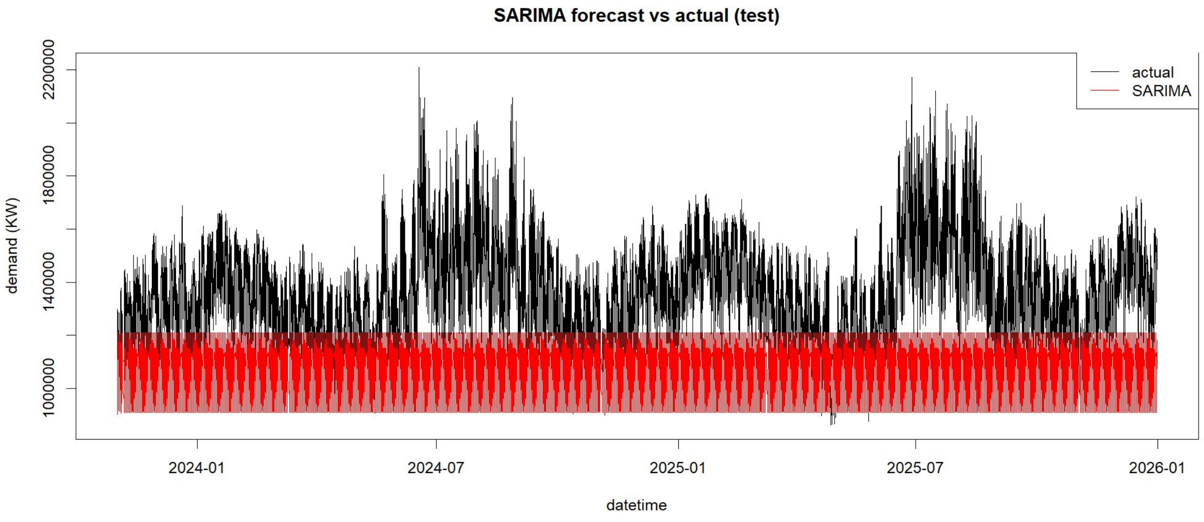
Autoregressive + 24-hour seasonal differencing on log(demand). No exogenous inputs.

Test RMSE: 337,752 KW

SARIMAX · SARIMA + apparent_temp as exogenous

Same structure as SARIMA plus apparent temperature as single exogenous regressor.

SARIMA one-step forecast (validation window)



Target: beat the ~100k-KW OLS line and cut SARIMA-family RMSE by ≥70%.

Three parallel approaches, one shared input

SHARED INPUT

Cleaned hourly matrix · 50,235 rows × 33 engineered features

COVID window dropped · log(demand) as response target

APPROACH 1

Diagnostic regression

- Start: OLS on raw features (adj $R^2 \approx 0.73$)
- VIF screen ⟶ keep collinear temp vars
- Box-Cox on response ⟶ log(demand)
- B-spline on apparent_temp (knots 30/50/70°F)
- Add holiday + heatwave flags
- Enhanced: + log(demand[t-1])

Split

Split: pre-2023 train / Jan–Oct 2023 val / Nov 2023+ test

APPROACH 2

Estimator benchmark

- Same cleaned matrix, sin/cos time encoding
- OLS · Ridge · LASSO · PCR · WLS
- Robust: Huber · LAD · LTS
- Time series: SARIMA · SARIMAX
- With / without VIF pruning
- With / without Lag-24 + Lag-168

Split

Split: pre-2023 train / Jan–Oct 2023 val / Nov 2023+ test

APPROACH 3

Deep learning benchmark

- Two-layer PyTorch LSTM
- Hidden size 200, sequence length 24
- Adam, learning rate 0.0005
- Early stopping at epoch 22
- First pass: weather-only (failed)
- Final: full 33-feature input set

Split

Split: same 3-way chronological as Approach 2

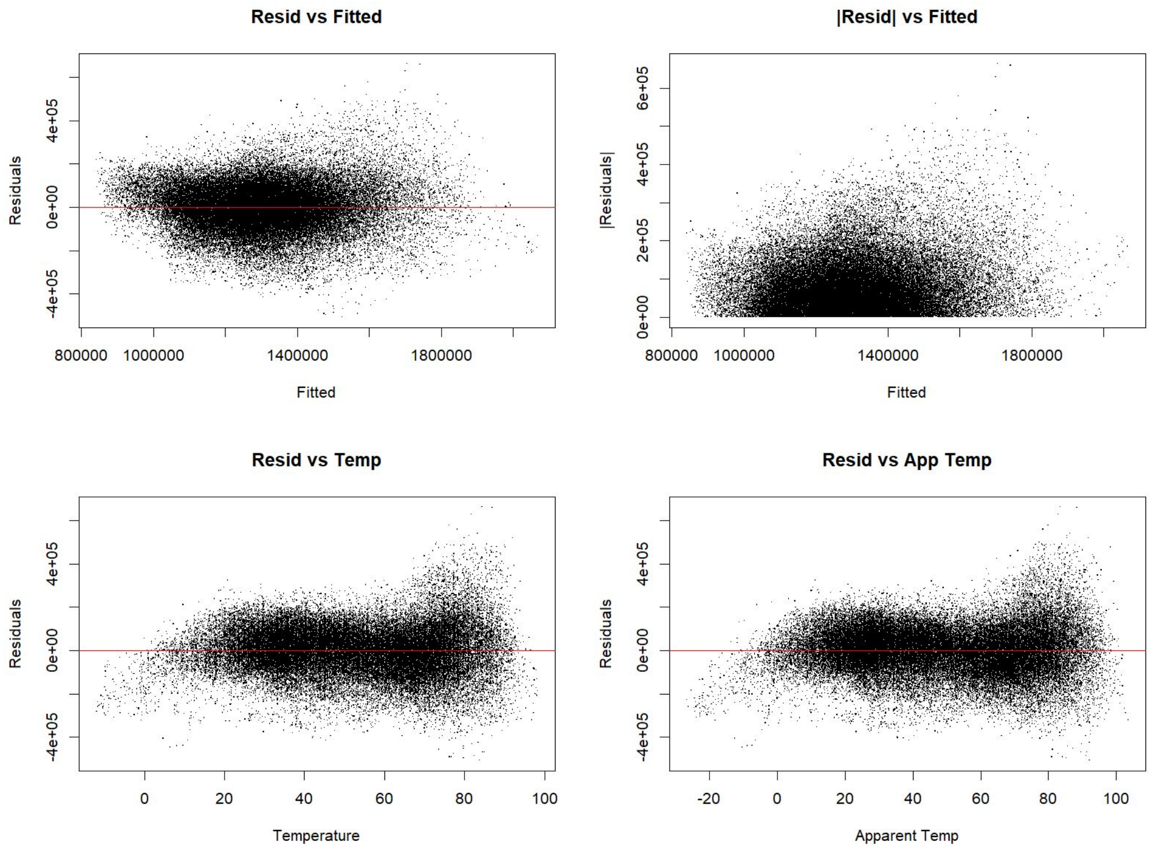
CONVERGE AT

Common test-window RMSE ⟶ see Slide 15 for the ranking

Approach 1 · OLS with raw features

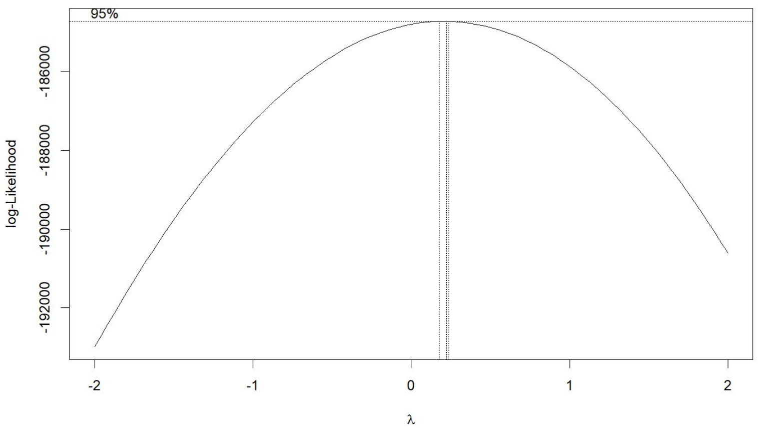
Where we started

- Linear regression on demand with raw weather, linear temperature, and calendar features.
- 19 weather predictors, no transforms.
- Adjusted $R^2 \approx 0.73$, test RMSE 120,328 KW.
- Residual vs fitted showed a clear non-linearity in temperature.
- QQ plot had heavy tails: the linear model is compromising between hot and cold extremes.

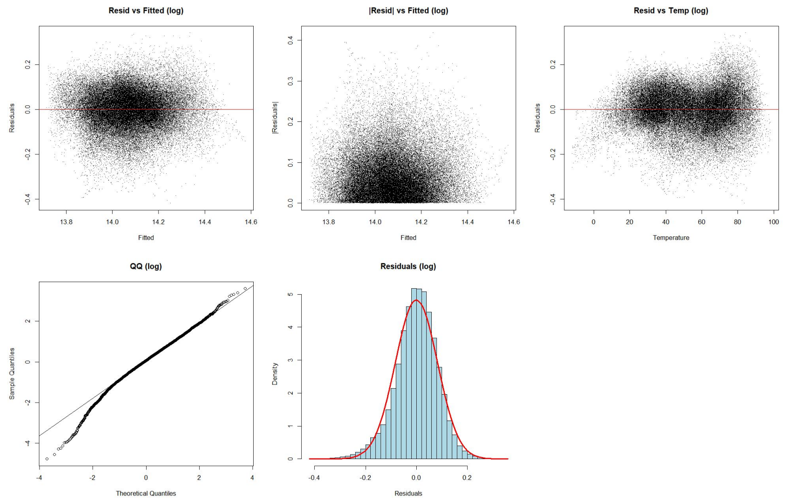


Approach 1 · Fixing response and temperature

Box-Cox profile



Post-transform diagnostics



WHY $\lambda = 0$ (LOG), NOT $\lambda = 0.3$

MLE point estimate for λ is 0.2-0.3, but the 95% CI is wide and the profile is nearly flat between $\lambda = 0$ and $\lambda = 0.3$. We chose $\lambda = 0$ because log is invertible cleanly, interpretable as a percentage change, and sits inside the CI. $\lambda = 0.3$ gives no meaningful RMSE gain and an awkward back-transform.

HELPED

log(demand): variance fan-out removed, heteroscedasticity test passes.

B-spline (30, 50, 70 °F): fixes the U-shape. Residuals flat across the full temp range.

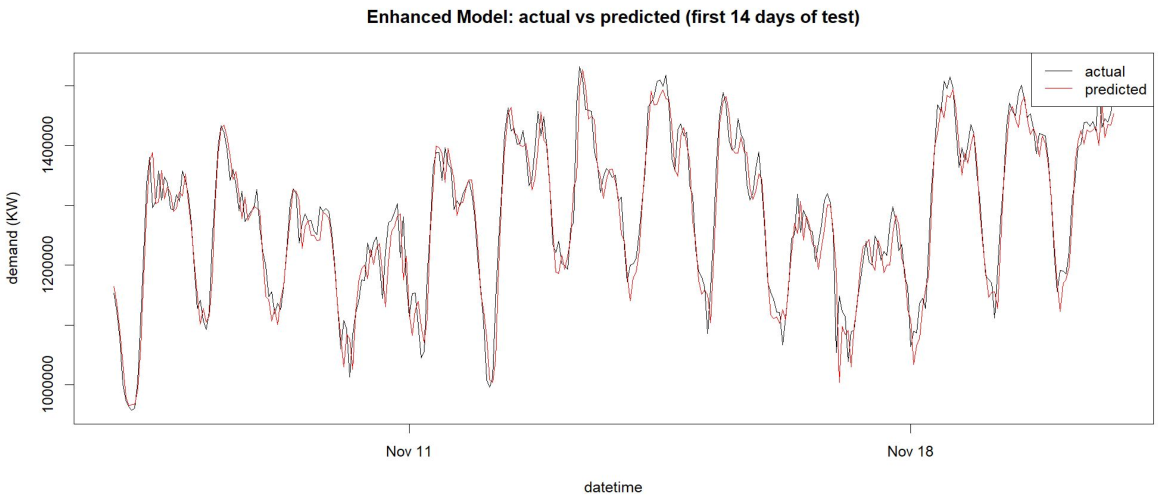
Holiday flag: captures the -11% demand drop on U.S. federal holidays.

Combined: adj R^2 0.727 → 0.752, test RMSE ≈ 120,500 KW held steady; the gain is in non-linear fit and cleaner residuals, not raw RMSE. Large RMSE reductions come later with lag-1.

DID NOT HELP

Higher-degree polynomials on humidity: VIF explodes, test RMSE unchanged.   Dropping collinear temp vars (VIF ≈ 410): costs 7% R^2 even though textbook advice says to drop them.

Approach 1 · Final model specification



FINAL MODEL · Approach 1

log(demand) =

β·bs(apparent_temp, knots = {30, 50, 70}, d = 3)

+ cloud_cover + VPD + shortwave_rad + soil_moisture

+ factor(hour) + factor(day-of-week) + factor(season)

+ holiday + heatwave

+ log(demand[t-1]) ⟵ Enhanced variant only

Heatwave flag (top-percentile apparent temp)

Explicit marker for extreme heat events. Lets the spline focus on normal-range response.

Holiday flag (6 US federal)

Biggest single indicator: demand drops ≈ 11% on federal holidays vs same weekday.

COVID HANDLING

Approach 1 does NOT use a COVID flag. The Mar 2020 – Jun 2021 window is dropped from the data before fitting, so there is nothing for a flag to capture. All diagnostics and plots on this slide and on slides 7-8 are computed on the post-drop sample (50,235 hours, 2019-2025 minus lockdown).

Primary (day-ahead, no lag)
120,544 KW, MAPE 7.42%, adj R^2 0.752

Enhanced (one-hour-ahead, actual lag-1)
44,772 KW, MAPE 2.53%, adj R^2 0.966

Recursive (multi-step, predicted lag)
121,048 KW, MAPE 7.48%

Approach 2 · Shrinkage and robust benchmarking

RELATIONSHIP TO APPROACH 1

Approach 2 starts from the same cleaned hourly matrix that Approach 1 used: COVID window already dropped, hour-24 rollovers fixed, comma formatting stripped, UTC normalized. Approach 1 does not add a COVID flag, and Approach 2 does not either — there is no COVID data in the training sample for a flag to explain.

Setup (what differs from A1)

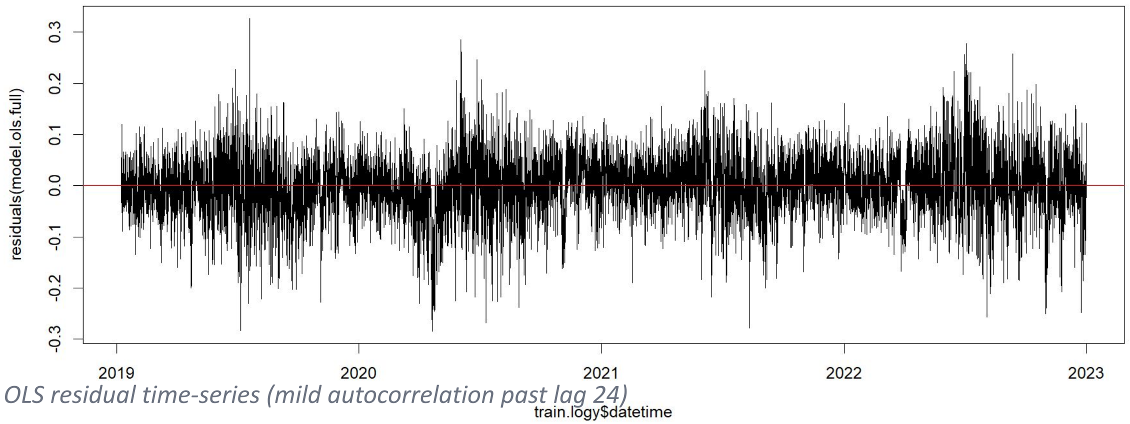
- Predictors include sin/cos time encoding (A1 uses factor levels).
- Temperature handled via cubic polynomial (A1 uses a B-spline).
- Split is three-way chronological (see right): 35,000+ train hours.
- Log response is shared with A1.
- OLS residual-vs-time shows mild autocorrelation past lag 24 (same shape as A1).

CHRONOLOGICAL SPLIT

Train: 2019-01 to 2022-12 (pre-2023, ~35,000 rows) , 57%/12%/31%

Validation: 2023-01 to 2023-10

Test: 2023-11 onward



OLS residual time-series (mild autocorrelation past lag 24)

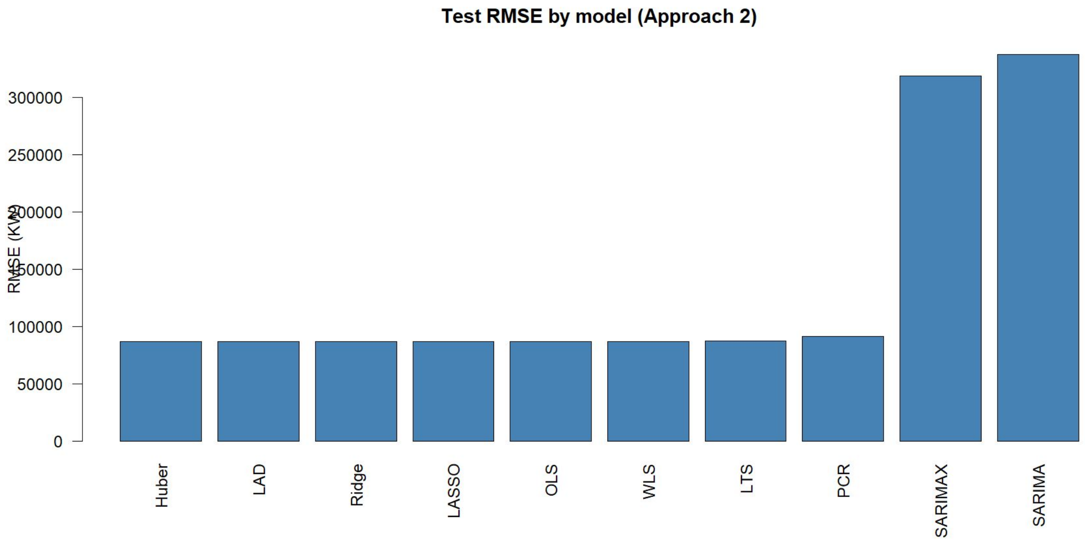
Approach 2 · Memory is the biggest lever

Incremental RMSE by feature group

Feature set	Test RMSE	Δ
Weather only	147,907 KW	baseline
+ Heatwave + Holiday flags	133,545 KW	-14,362
+ Lag24 + Lag168	**86,748 KW**	**-46,800**

Lag-24 + Lag-168 alone drop RMSE ~35%. Temporal memory beats fancy estimators.

Test RMSE across all Phase-2 models



Approach 2 · All linear estimators land together

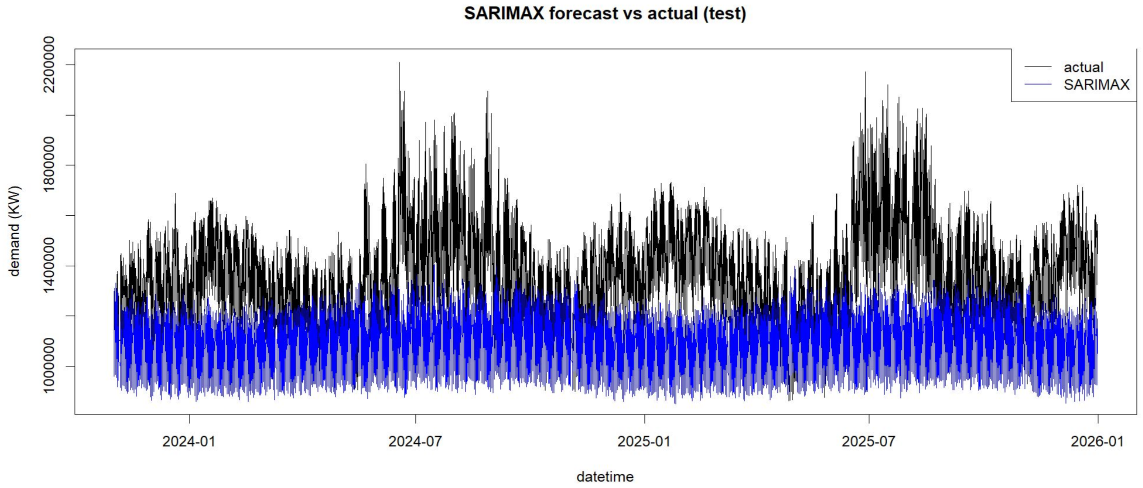
With flags, every shrinkage and robust estimator converges:

OLS	**133,604 KW**	**Ridge**	**133,614 KW**
LASSO	133,603 KW	**PCR**	150,391 KW
Huber	134,211 KW	**LAD**	135,684 KW
LTS	136,846 KW		

Why they tie

Before flags, robust methods (LAD) beat OLS because they implicitly down-weight holidays and extreme-heat outliers. Once those events are flagged, there is nothing left for robust methods to clean up. PCR was the only clear loser.

SARIMAX forecast (for contrast)



SARIMA 337,752 · SARIMAX 318,583 · every linear + flags model ≈ 133,000. Weather + flags wins on its own.

Approach 3 · First LSTM attempt fails

Architecture

- Two-layer LSTM, hidden size 200, sequence length 24.
- Inputs: weather features only.
- No calendar, no lags, no COVID flag.
- Standard train / validation / test split on the cleaned hourly series.

TEST RMSE

126,480 KW

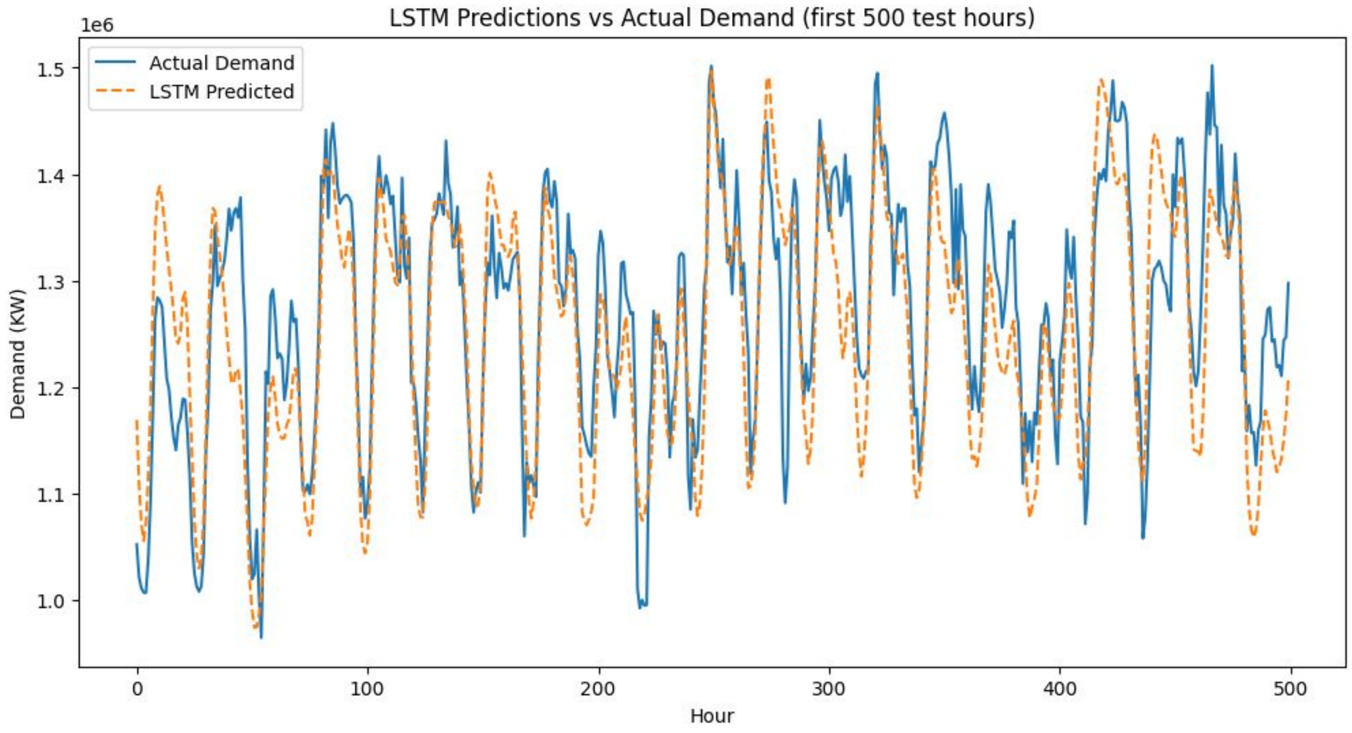
Weather-only LSTM

+46% worse than OLS (86,700 KW)

Networks cannot invent features.

Without calendar structure or temporal lags, the LSTM has no signal that a Sunday at 2 a.m. differs from a Wednesday at 2 p.m. beyond current weather.

Approach 3 · Full-feature LSTM is the best single model



TEST RMSE

78,432 KW

Test MAE: 62,514 KW

Split

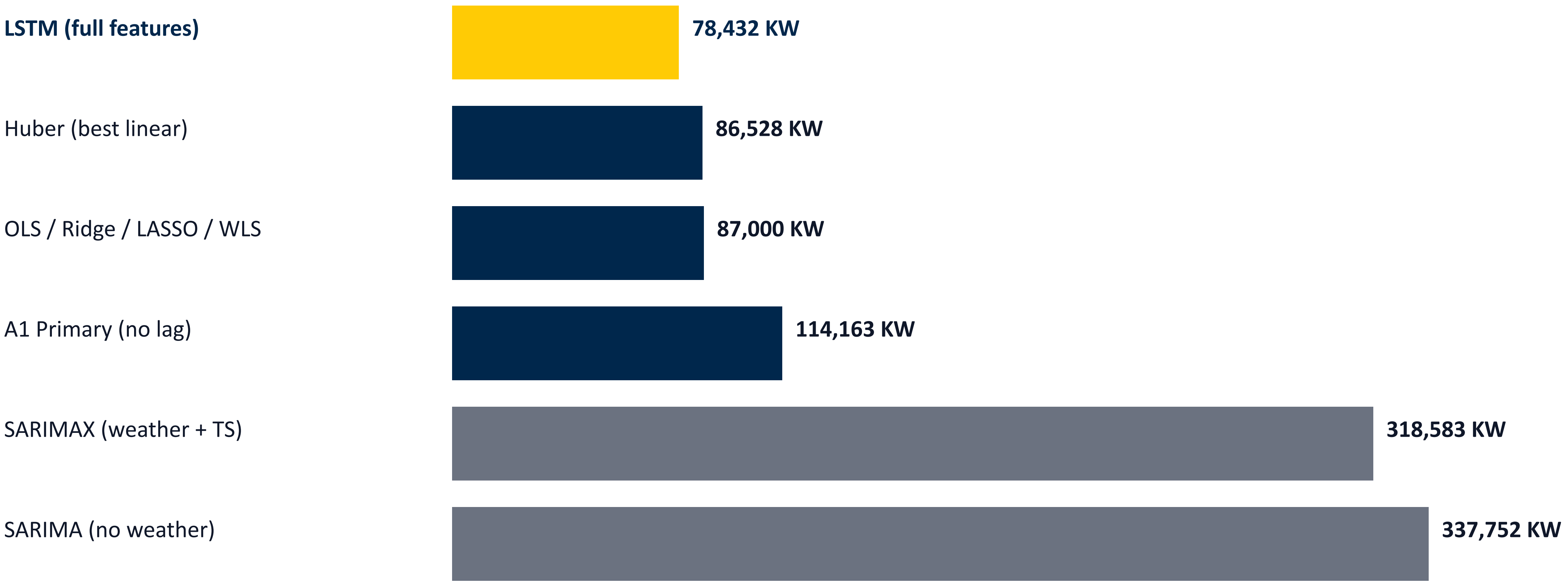
34,882 train · 7,294 val · 18,883 test

Features

Weather · calendar · holiday · heatwave · Lag24 · Lag168

Gain over Huber (best linear): about 10%. The network is exploiting non-linear feature interactions the linear family could not.

All models ranked by test RMSE



LSTM with full features wins by ~10% over the best linear model.

What 78k KW of error actually costs

LSTM TEST RMSE

78,432 KW

about 6.0% of mean demand

HOURLY COST EQUIVALENT

≈ $3,980 / hour

at PJM 2025 LMP of $50.73/MWh → about $34.9M per year of gross energy-market exposure (upper bound).

Sources: Monitoring Analytics 2025 State of the Market for PJM; PJM Load Forecasting manuals.

VERSUS THE FIELD

▲ vs SARIMA baseline (337,752 KW)

saves ~$115M / yr

▼ vs PJM production forecast (~3% MAPE)

costs ~$17.5M / yr more

VERDICT

Our best model sits at the upper edge of the PJM MAPE band. It would not replace a production forecast without losing money.

The value here is the side-by-side comparison of classical and deep models on one utility's data.

Where this work falls short, and thank you.

Collinearity

Temp vs apparent temp VIF > 400. Individual coefficients cannot be read.

Spring coverage

Approach 1 PICP 0.893 in spring; hot-tail PICP 0.882 above 90 °F.

Approach 2 PICP 0.916; hot-tail PICP 0.675

Residual-variance assumption is stationary; Ohio spring is not.

Heat extremes

Approach 2 summer PICP drops from 0.892 to 0.675 above 90 °F.

One-step horizon

Enhanced model uses an observed lag. A real day-ahead deployment would need a forecasted lag and recursive evaluation.

Thank you · Questions?

Hamzah Harsani & Maximilian Ermler · IOE 562 Winter 2026